Assignment No :- 9

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Q.1] Selection Sort Que.

Problem

Sort an array of integers in non-decreasing order using Selection sort

a) Input 1 :

[18, 22, 20, 19, 21]

Output :

[18, 19, 20, 21, 22]

~~{18, 19, 20, 21, 22}~~
~~unsorted~~
~~min = 18~~
~~{18, 19, 20, 21, 222}~~
~~sort~~ ~~unsort~~
~~min~~

{18, 22, 20, 19, 21}
unsorted
min = 18
{18, 22, 20, 19, 21}
sort unsort
min = 19

{18, 19, 20, 22, 21}
sort unsort
min = 20

{18, 19, 20, 22, 21}
sort unsort
min = 21

{18, 19, 20, 21, 22}
sort unsort

{18, 19, 20, 21, 22}
sorted

b) Input 2 :- {34, 12, 56, 45, 23} → {12, 23, 34, 40, 56}

{34, 12, 56, 45, 23}
unsort
min = 12

{12, 34, 56, 45, 23}
sort unsort
min = 23

{12, 23, 56, 45, 34}
sort unsort
min = 34

{12, 23, 34, 45, 56}
sort unsort
min = 45

{12, 23, 34, 45, 56}
sort unsort

{12, 23, 34, 45, 56}
sorted.

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Q.2] Bubble sort :-

Given an array, sort in non-decreasing order using Bubble sort.

a) Input : i

$\{30, 25, 27, 35, 29\} \rightarrow [25, 27, 29, 30, 35]$

$\{\underbrace{30, 25, 27, 35, 29}_{\text{unsort}}\}$

Step 1]

$= \{30, 25, 27, 35, 29\}$

$= \{25, 30, 27, 35, 29\}$

$= \{25, 27, 30, 35, 29\}$

$= \{25, 27, 30, 29, 35\}$

Step2] $\{\underbrace{25, 27, 30, 29}_{\text{Unsort}}, \underbrace{35}_{\text{sort}}\}$

$= \{25, 27, 30, 29, 35\}$

$= \{\underbrace{25, 27, 29}_{\text{Unsort}}, \underbrace{30, 35}_{\text{sort}}\}$

Step3]

$\{22, 27, 29, 30, 35\}$

$\{\underbrace{22, 27}_{\text{Unsort}}, \underbrace{29, 30, 35}_{\text{sort}}\}$

Step4]

$\{22, 27, 29, 30, 35\}$

$\{\underbrace{22}_{\text{unsort}}, \underbrace{27, 29, 30, 35}_{\text{sort}}\}$

$\{\underbrace{22, 27, 29, 30, 35}_{\text{sorted}}\}$

Input 2 :-

[15, 10, 25, 40, 30] → [10, 15, 25, 30, 40]

{15, 10, 25, 40, 30}
unsort

Step 1 :- {15, 10, 25, 40, 30}

= {10, 15, 25, 40, 30}

= {10, 15, 25, 30, 40}
Unsort Sort

Step 2 :-

= {10, 15, 25, 30, 40}

= {10, 15, 25, 30, 40}
Unsort Sort

Step 3 :- {10, 15, 25, 30, 40}

= {10, 15, 25, 30, 40}

= {10, 15, 25, 30, 40}
Unsort sort

Step 4 :-

{10, 15, 25, 30, 40}

= {10, 15, 25, 30, 40}
Unsort Sort

{10, 15, 25, 30, 40}
Sorted.

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Q.3] Insertion sort :-

Input :) [18, 22, 20, 19, 21] → [18, 19, 20, 21, 22]

{18, 22, 20, 19, 21}

1] {18, 22, 20, 19, 21}
sorted Unsort

18 > 22

{18, 22, 20, 19, 21}
sort Unsort

2] {18, 22, 20, 19, 21}

22 > 20
shift

{18, 22, 22, 19, 21}

18 > 20

{18, 20, 22, 19, 21}
sort unsort

3] {18, 20, 22, 19, 21}

22 > 19
shift

{18, 20, 22, 22, 21}

20 > 19 shift

{18, 20, 20, 22, 21}

18 > 19
×

{18, 19, 20, 22, 21}
sort Unsort

4] {18, 19, 20, 21, 21}

22 > 21
shift

{18, 19, 20, 22, 22}

20 > 21
×

{18, 19, 20, 21, 22}
sorted

Input 2 :- [34, 12, 56, 45, 23] → [12, 23, 34, 45, 56]

[34, 12, 56, 45, 23]
Unsort

1] [34, 12, 56, 45, 23]
Sort Unsort

34 > 12
shift

[34, 34, 56, 45, 23]

[12, 34, 56, 45, 23]
Sort Unsort

2] [12, 34, 56, 45, 23]

34 > 56
12 > 56

[12, 34, 56, 45, 23]
Sort Unsort

3] [12, 34, 56, 45, 23]

56 > 45
shift

[12, 34, 56, 56, 23]

34 > 45

[12, 34, 45, 56, 23]
Sort Unsort

4] [12, 34, 45, 56, 23]

56 > 23
shift

[12, 34, 45, 56, 56]

45 > 23
shift

[12, 34, 45, 45, 56]

34 > 23
shift

[12, 34, 34, 45, 56]

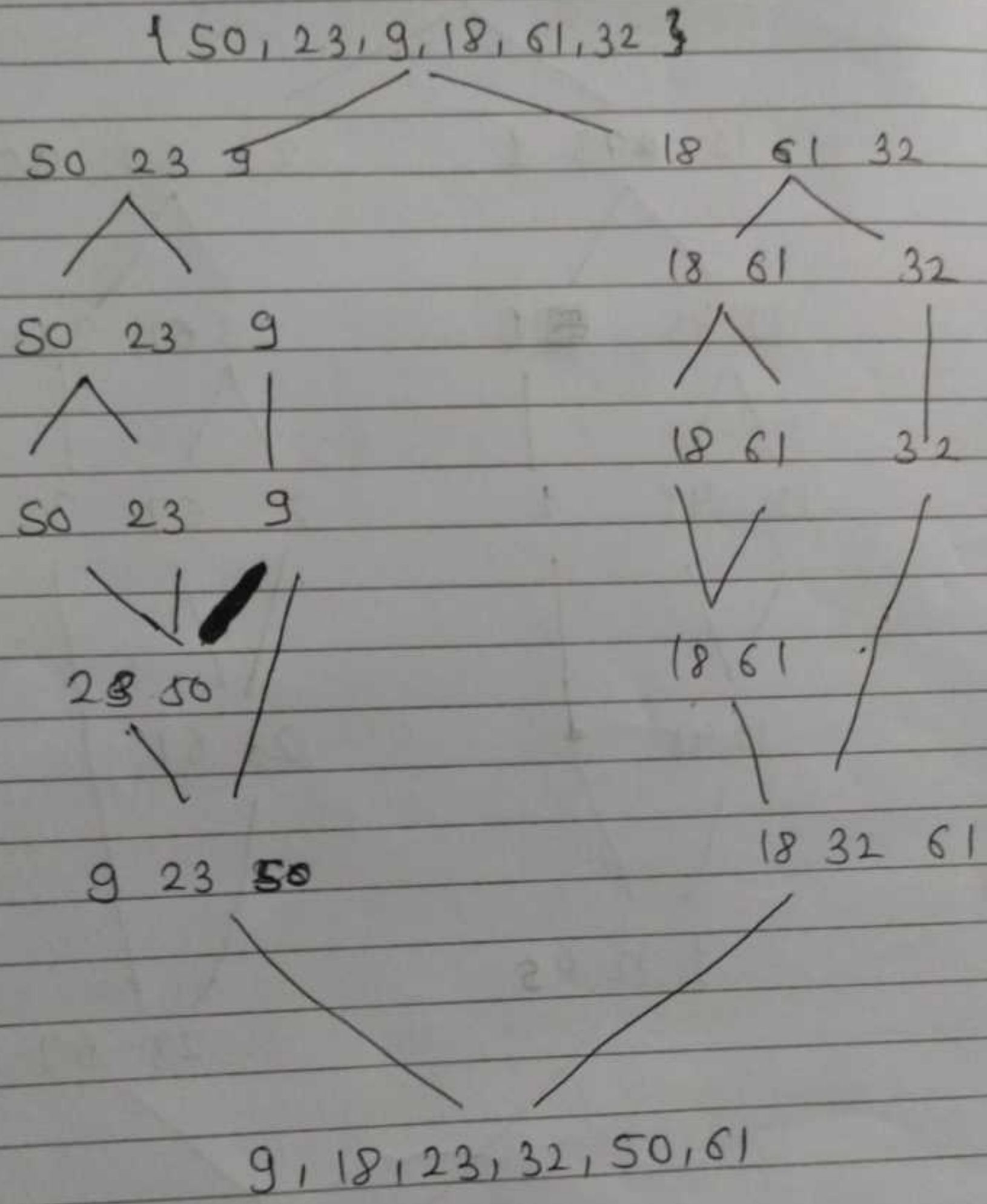
12 > 23

[12, 23, 34, 45, 56]
sorted.

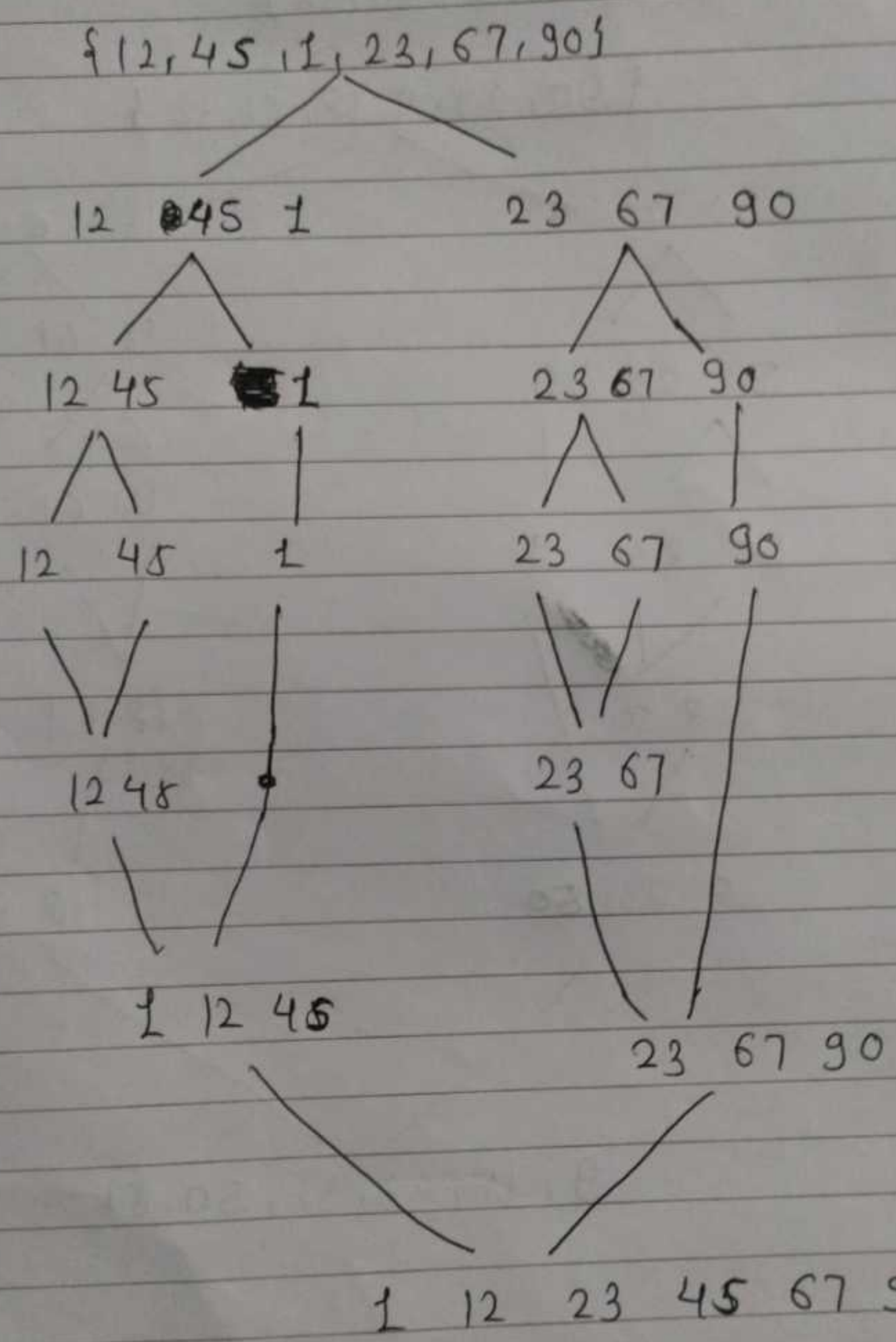
Q.4] Merge Sort

Input :
[50, 23, 9, 18, 61, 32]

→



Input 2: [12, 45, 1, 23, 67, 90]



{85, 95}

95}

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Q.5] Quick Sort

Input 1 :
[75, 85, 55, 95, 65]

Pivot = 75 {55, 65, 75, 85, 95}

0	1	2	3	4
65	55	75	85	95

Smaller Bigger

{65, 55} {85, 95}

Pivot = 65 {55, 65} Pivot = 85 {85, 95}

0	1
55	65

Smaller

0	1
85	95

Bigger

{55} {95}

1) Choose a pivot
2) Place the pivot at its correct position
→ ~~replace~~ put small element at left side of pivot

→ put high element at right side of pivot.

4) Repeat left part and right part.

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Q:5) Input 2)- [42, 56, 12, 67, 34]

{42, 56, 12, 67, 34}

Pivot = 42

{12, 34, 42, 56, 67}
0 1 2 3 4

34	12	42	56	67
0	1	2	3	4

Smaller — Bigger

{55, 67}

{34, 12}

Pivot = 55

55	67
0	1

{55, 67}

Bigger

{67}

Pivot = 34 {12, 34}

12	34
0	1

Smaller

{12}